

skipping Base Cases

$$\text{Inductive Hypothesis: } \sum_{i=1}^n \frac{1}{i(i+1)(i+2)} = \frac{n^2 + 3n}{4(n+1)(n+2)}$$

6.3) 2) Induction Step: we have to prove that

$$\sum_{i=1}^{n+1} \frac{1}{i(i+1)(i+2)} = \frac{(n+1)^2 + 3(n+1)}{4(n+2)(n+3)}$$

By Recursive Definition

$$\begin{aligned} &= \sum_{i=1}^n \frac{1}{i(i+1)(i+2)} + \frac{1}{(n+1)(n+2)(n+3)} \\ &= \frac{n^2 + 3n}{4(n+1)(n+2)} + \frac{1}{(n+1)(n+2)(n+3)} \end{aligned}$$

$$\begin{aligned} &= \frac{(n+1)[(n+1)^2 + 3(n+1)]}{4(n+1)(n+2)(n+3)} \end{aligned}$$

($\therefore$ multiplying & dividing by $(n+1)$ to have $n+1$ in denominator)

($\therefore$ By Inductive Hypothesis)

$$= \frac{(n+3)(n^2 + 3n) + 4}{4(n+1)(n+2)(n+3)}$$

$$= \frac{n^3 + 6n^2 + 9n + 4}{4(n+1)(n+2)(n+3)}$$

$$= \frac{(n+1)[n^2 + 2n + 1 + 3n + 3]}{4(n+1)(n+2)(n+3)}$$

$$= \frac{n^3 + 5n^2 + 4n + n^2 + 5n + 4}{4(n+1)(n+2)(n+3)}$$

Go back

6.3) 6) Inductive Hypothesis: $\sum_{i=1}^n \frac{1}{i^2} \leq 2 - \frac{1}{n}$ — ①

Prove that $\sum_{i=1}^{n+1} \frac{1}{i^2} \leq 2 - \frac{1}{(n+1)}$ — ②

By Recursive Definition:

$$\sum_{i=1}^{n+1} \frac{1}{i^2}$$

$$= \sum_{i=1}^n \frac{1}{i^2} + \frac{1}{(n+1)^2}$$

($\therefore$ by ①)

$$\leq 2 - \frac{1}{n} + \frac{1}{(n+1)^2}$$

$$= 2 - \frac{n}{n(n+1)^2} + \frac{1}{(n+1)^2}$$

$$2 - \left[\frac{(n+1)^2 - n}{n(n+1)^2} \right]$$

$$2 - \left[\frac{n^2 + n + 1}{n(n+1)^2} \right]$$

&

$$\begin{aligned} &= 2 - \frac{n^2 + n + 1}{n(n+1)^2} \\ &= 2 - \frac{n(n+1 + \frac{1}{n})}{n(n+1)^2} \\ &= 2 - \frac{n+1 + \frac{1}{n}}{(n+1)^2} \end{aligned}$$

$$< 2 - \frac{1}{(n+1)^2} \quad (\because \text{as } n+1 + \frac{1}{n} > 1)$$

as $n \geq 1$

↓
multiply by -1 & add 2
to get required inequality

6.3) 7) (a) Inductive Hypothesis: $\sum_{i=0}^n (a_i + b_i) = \sum_{i=0}^n a_i + \sum_{i=0}^n b_i$ — ①

Prove that $\sum_{i=0}^{n+1} (a_i + b_i) = \sum_{i=0}^{n+1} a_i + \sum_{i=0}^{n+1} b_i$

$$\begin{aligned} \text{By Recursive Def L.H.S.} &= \sum_{i=0}^n (a_i + b_i) + a_{n+1} + b_{n+1} \\ &= \sum_{i=0}^n a_i + a_{n+1} + \sum_{i=0}^n b_i + b_{n+1} \quad (\because \text{by } ①) \\ &= \sum_{i=0}^{n+1} a_i + \sum_{i=0}^{n+1} b_i = \text{R.H.S.} \end{aligned}$$

(b) Inductive Hypothesis: $c \cdot \sum_{i=0}^n a_i = \sum_{i=0}^n (c \cdot a_i)$ — ①

Prove that $c \cdot \sum_{i=0}^{n+1} a_i = \sum_{i=0}^{n+1} (c \cdot a_i)$

$$\begin{aligned} \text{L.H.S.} &= c \left[\sum_{i=0}^n a_i + a_{n+1} \right] \quad \because \text{By Recursive Def.} \\ &= c \sum_{i=0}^n a_i + c \cdot a_{n+1} \\ &= \sum_{i=0}^n (c \cdot a_i) + c \cdot a_{n+1} \quad \because \text{By } ① \\ &= \sum_{i=0}^{n+1} (c \cdot a_i) = \text{R.H.S.} \end{aligned}$$

6.3) 12) (a) if $2^n \geq n+1$ then $2^n > n$

so Inductive Hypothesis: $2^n \geq n+1$ — ①

Inductive step: $2^{n+1} = 2^n \cdot 2$

$$\geq (n+1) \cdot 2 \quad \because \text{By } ①$$

$$= 2n+2$$

$$\geq n+2 \quad \because 2n \geq n$$

6.3) 13) $k \in \mathbb{Z}^+$ — ①

Inductive Hypothesis: $(k^2+n)! \geq k^{2^n}$ — ②

Prove that $(k^2+n+1)! \geq k^{2^{n+1}}$ — ③

$$\begin{aligned} \text{Induction step: } (k^2+n+1)! &= (k^2+n+1)(k^2+n)! \quad \because \text{By Recursive Def.} \\ &\geq (k^2+n+1)(k^{2^n}) \quad \because \text{By } ② \\ &= k^{2^n+2} + n k^{2^n} + k^{2^n} \quad \because k \geq 1 \text{ by } ① \\ &\geq k^{2^n+2} \\ &= k^{2^{n+1}} \end{aligned}$$

6.3) 14) simply we do induction on n by taking m to be arbitrary: $(a^m)^{n+1} = (a^m)^n \cdot a^m$ $\therefore$ By Recursive Def
 $= a^{mn} \cdot a^m$ $\therefore$ By inductive Hypothesis
 $= a^{mn+m}$ $\therefore$ Theorem on pg 297
 $= a^{m(n+1)}$

6.3) 16) $a_0 = 2$, $a_1 = 2^2 = 4$, $a_2 = 16$, $a_3 = 256$...
 $\downarrow 2^{2^0}$ $\downarrow 2^{2^1}$ $\downarrow 2^{2^2}$ $\downarrow 2^{2^3}$...

Conjecture: $a_n = 2^{2^n}$

Proof: Base case: $a_0 = 2 = 2^1 = 2^{2^0}$

Inductive Hypothesis $a_n = 2^{2^n}$ — ①

Induction step: $a_{n+1} = (a_n)^2$

$$= (2^{2^n})^2 \quad \therefore \text{By } ①$$

$$= 2^{2^n \cdot 2} = 2^{2^{n+1}}$$

6.3) 19) (a) Inductive Hyp: $\sum_{k=0}^n \binom{n}{k} = 2^n$

Induction step:

$$\sum_{k=0}^{n+1} \binom{n+1}{k} = 2^{n+1}$$

$$\text{L.H.S} = \binom{n+1}{0} + \binom{n+1}{1} + \dots + \binom{n+1}{n} + \binom{n+1}{n+1}$$

$$= 1 + \binom{n}{0} + \binom{n}{1} + \binom{n}{1} + \binom{n}{2} + \dots + \binom{n}{n-1} + \binom{n}{n} + 1$$

$$= 2 + 1 + 2\binom{n}{1} + 2\binom{n}{2} + \dots + 2\binom{n}{n-1} + 1$$

$$= 2\binom{n}{0} + 2\binom{n}{1} + 2\binom{n}{2} + \dots + 2\binom{n}{n-1} + 2\binom{n}{n}$$

$$= 2 \left[\binom{n}{0} + \binom{n}{1} + \dots + \binom{n}{n-1} + \binom{n}{n} \right]$$

$$= 2 \sum_{k=0}^n \binom{n}{k}$$

$$= 2 \cdot 2^n = 2^{n+1} = \text{R.H.S}$$

A combinatorial Proof can also be given as L.H.S just counts the total number of subsets of set A with $|A| = n$. Another way is to plug $x=y=1$ in Binomial Theorem in exercise 18 (ol)